

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236 B

Total Parameters

21 B

Activated per Token

128 K tokens

Context Length

8.1 T

tokens
Pretraining corpus

The LLM Scaling Dilemma: Performance vs. Cost & Inference Speed

The Problem

- ▶ Larger dense models improve capability, but training costs scale quadratically with parameter count.
- ▶ Autoregressive decoding stores full K/V pairs per head per token → KV cache grows to GB scale.
- ▶ KV cache bottleneck limits batch size and throughput at inference time.
- ▶ Naive model scaling is neither economically nor environmentally sustainable.

The Solution

- ▶ Mixture-of-Experts (MoE) activates only a fraction of parameters per token → far cheaper training.
- ▶ Low-rank KV compression can shrink the cache without sacrificing attention expressiveness.
- ▶ DeepSeek-V2 combines both: MLA for attention + DeepSeekMoE for FFN layers.
- ▶ Result: 21B activated out of 236B total — intelligence of a 70B+ model at a fraction of the cost.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in Each Transformer Block

Transformer Block Components

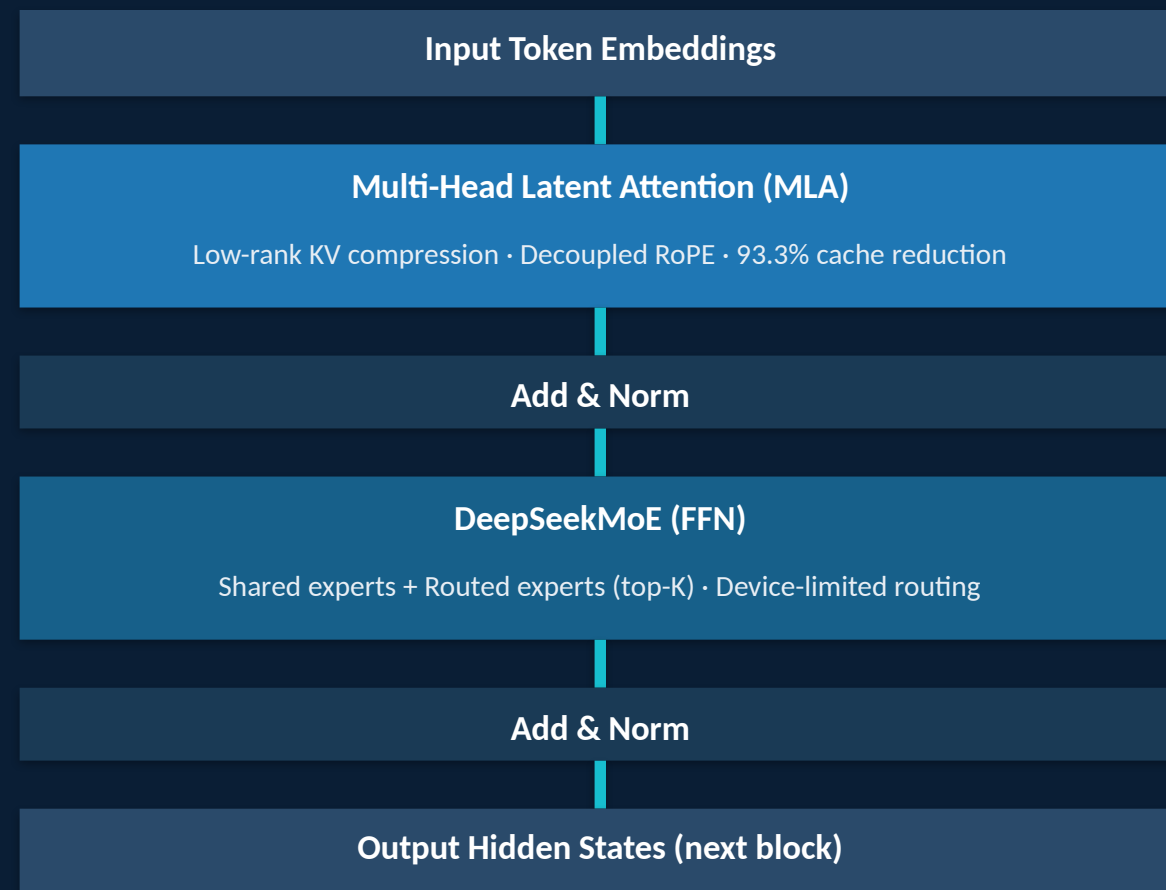
- ▶ Each block contains: Multi-Head Latent Attention (MLA) replacing standard MHA, and DeepSeekMoE replacing the dense FFN.
- ▶ Both innovations target orthogonal bottlenecks: memory (MLA) and compute/cost (MoE).

MLA Layer

- ▶ Projects queries, keys, and values into low-rank latent space.
- ▶ Caches compact latent vectors instead of full K/V matrices.
- ▶ Decoupled RoPE applied separately to positional components.

DeepSeekMoE Layer

- ▶ N_s shared experts always active + N_r routed experts (top- K_r selected).
- ▶ Fine-grained expert segmentation for higher specialization.
- ▶ Device-limited routing reduces inter-node communication.



Reference: Figure 1 — DeepSeek-V2 Architecture Diagram

Multi-Head Latent Attention: 93.3% KV Cache Reduction with Better Performance

Standard MHA caches full key and value vectors per head per token — this creates a per-token memory cost that grows with both heads and sequence length. MLA instead compresses K and V jointly into a low-rank latent vector c_t^{KV} via learned down-projection. At inference, full K/V are reconstructed on-the-fly; only the compact latent is stored.

MHA

Multi-Head Attention

- ▶ Caches: full K & V per head
- ▶ Cache size: $n_h \times d_h$ per token
- ▶ Performance: Strong baseline
- ▶ Heads: all independent

GQA

Grouped-Query Attn

- ▶ Caches: shared K,V per group
- ▶ Cache size: $n_g \times d_h$ per token
- ▶ Performance: small degradation
- ▶ Groups: $G < n_h$

MQA

Multi-Query Attn

- ▶ Caches: single K,V pair
- ▶ Cache size: $1 \times d_h$ per token
- ▶ Performance: noticeable drop
- ▶ Extreme cache reduction

MLA ★

Multi-Head Latent Attn

- ▶ Caches: latent vector c_t^{KV} only
- ▶ Cache size: ~6.25% of MHA
- ▶ Performance: BETTER than MHA
- ▶ Decoupled RoPE for positions

Reference: Figure 2 — Attention mechanism comparison (MHA / GQA / MQA / MLA)

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Two Expert Types

- ▶ Shared experts (Ns): always activated — capture universal, cross-domain knowledge.
- ▶ Routed experts (Nr): sparsely activated via top-Kr gating — learn specialised patterns.

Fine-Grained Segmentation

- ▶ Experts are smaller but more numerous than standard MoE, increasing total specialization capacity.
- ▶ Each expert becomes more focused, improving routing efficiency.

Device-Limited Routing

- ▶ Each token is routed to experts on at most M devices.
- ▶ Dramatically reduces cross-node all-to-all communication overhead at scale.

Training Stability

- ▶ Auxiliary balance losses prevent expert collapse and ensure uniform utilisation.
- ▶ Token-dropping strategy handles overflow without affecting inference quality.



42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

Relative Efficiency Gains: DeepSeek-V2 vs. DeepSeek 67B (Dense Baseline)

-42.5%

Training Cost

Reduction vs. DeepSeek 67B

-93.3%

KV Cache Size

Reduction (MLA vs. MHA)

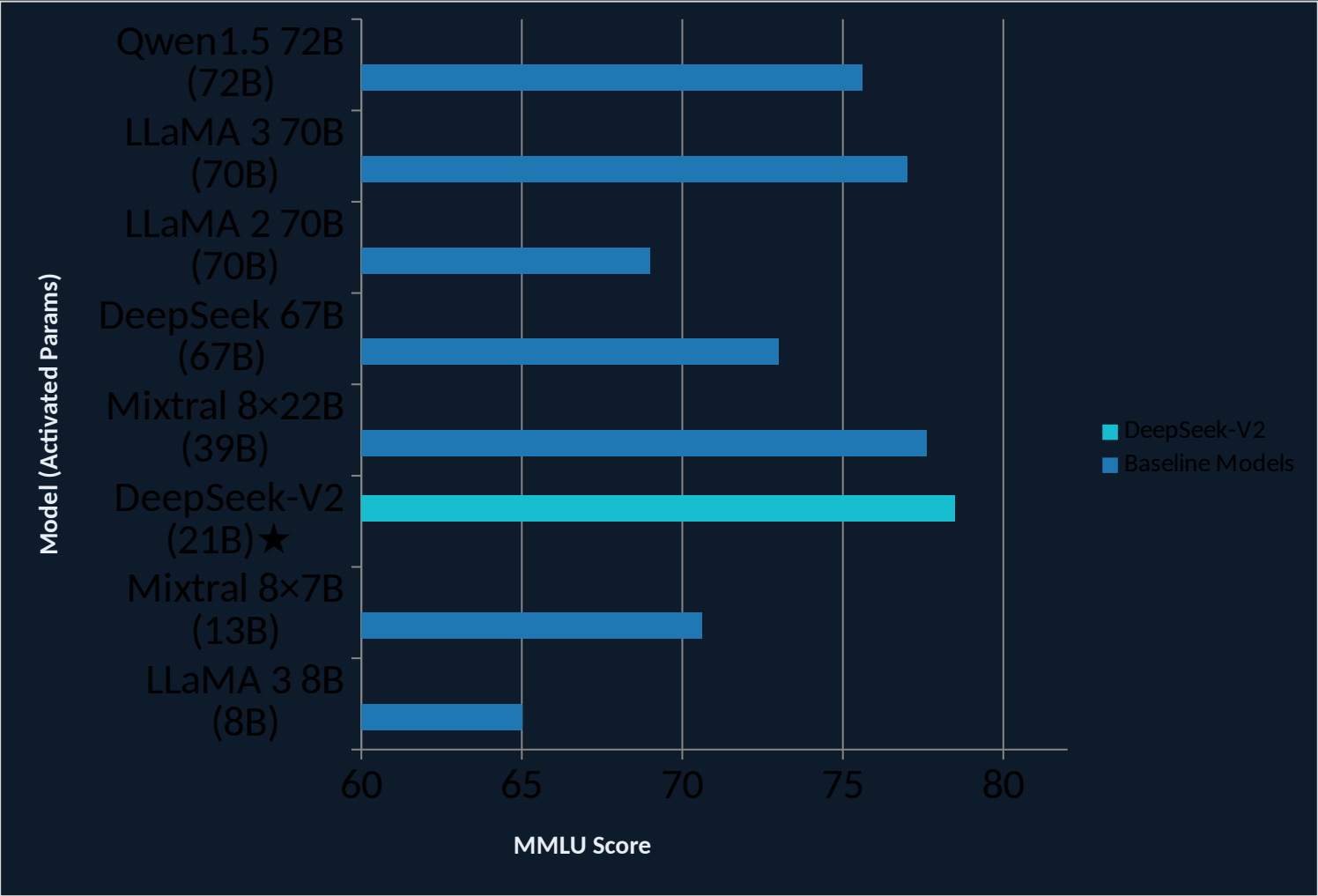
5.76×

Max Gen. Throughput

Higher than DeepSeek 67B

Metric	DeepSeek 67B (Dense)	DeepSeek-V2 (MoE)	Improvement
Training Cost	Baseline (100%)	57.5% of baseline	42.5% lower ↓
KV Cache	Baseline (100%)	6.7% of baseline	93.3% smaller ↓
Max Gen. Throughput	Baseline (1×	5.76× of baseline	5.76× faster ↑
Activated Params	67 B	21 B	68.7% fewer

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters — Pareto-Optimal



Reference: Figure 3 — MMLU vs. Activated Parameters

MMLU Scores vs. Activated Parameters

Model	Act. Params	MMLU
DeepSeek-V2	21 B	78.5
LLaMA 3 70B	70 B	77.0
Mixtral 8×22B	39 B	77.6
Qwen1.5 72B	72 B	75.6
DeepSeek 67B	67 B	73.0
Mixtral 8×7B	13 B	70.6
LLaMA 2 70B	70 B	69.0
LLaMA 3 8B	8 B	65.0

Pareto-Optimal Positioning

Highest MMLU (78.5) among open-source models while activating only 21B params — far fewer than all 70B+ competitors.

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chat Benchmarks

Alignment Pipeline: Pretrained base → Supervised Fine-Tuning (SFT) on 1.5 M conversational sessions → Reinforcement Learning via Group Relative Policy Optimization (GRPO) → DeepSeek-V2 Chat (RL)

38.9%

AlpacaEval 2.0

Length-controlled win rate
vs. GPT-4 reference

8.97 / 10

MT-Bench

Multi-turn instruction
following (overall)

7.91 / 10

AlignBench

Open-ended Chinese
conversation quality

- ▶ AlignBench (Chinese): DeepSeek-V2 Chat (RL) outperforms ALL open-source models and MOST closed-source models, including GPT-3.5-series systems.
- ▶ MT-Bench (8.97): competitive with leading proprietary instruction-following models.
- ▶ AlpacaEval 2.0 (38.9% LC win rate): strong open-ended English conversational quality.
- ▶ GRPO enables effective RL alignment without reward model instability, making the training pipeline practical at scale.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

MLA Supersedes MHA

Low-rank KV compression achieves better performance than standard Multi-Head Attention while cutting the KV cache by 93.3%. MLA is a strictly better alternative to GQA/MQA for inference efficiency.

DeepSeekMoE Enables Economical Training

Fine-grained shared + routed expert design trains a 236B-parameter model at 42.5% lower cost than a 67B dense model — demonstrating that MoE at scale is both practical and economical.

21B Activated Params Match 70B+ Dense Models

DeepSeek-V2 achieves 78.5 MMLU, topping all open-source models, while activating only 21B out of 236B parameters — outperforming LLaMA 3 70B (77.0) and Qwen1.5 72B (75.6).

5.76× Generation Throughput Enables Real Deployment

Substantially higher throughput over the dense DeepSeek 67B baseline means DeepSeek-V2 is not just research-efficient — it is practical for large-scale inference serving.

Strongest Open-Source Chat Model in Chinese

DeepSeek-V2 Chat (RL) surpasses all open-source and most closed-source systems on AlignBench (7.91), demonstrating strong multilingual alignment via SFT + GRPO post-training.